

Conceptual Answers – OFI Feature Engineering

Sushal Reddy Sabbella

1. Why measure OFI at multiple levels?

Measuring Order Flow Imbalance (OFI) at multiple levels of the order book provides a richer view of market dynamics. While the best bid and ask prices (Level 0) are crucial for immediate trading decisions, deeper levels (e.g., Levels 1-10) contain hidden liquidity and passive interest that may affect future price impact. Large institutional orders often interact with multiple layers of the book. Therefore, multi-level OFI captures both immediate and potential liquidity pressure, leading to more robust signals for short-term price prediction.

2. Why use LASSO instead of OLS in cross-asset modeling?

LASSO (Least Absolute Shrinkage and Selection Operator) is preferred over OLS (Ordinary Least Squares) in cross-asset settings due to high dimensionality and potential multicollinearity. In such models, each asset's return may be influenced by many other assets' OFIs. LASSO adds an L1 penalty to the regression, enforcing sparsity and variable selection, which helps isolate the most significant cross-impact relationships while reducing overfitting and improving interpretability.

3. Why is OFI a better predictor than trade volume?

Trade volume is an undirected measure—it tells us how much trading occurred, but not whether buyers or sellers were dominant. In contrast, OFI directly measures the imbalance between supply and demand by comparing the change in bid and ask queue sizes. This directional signal provides more informative insight into market pressure and is empirically shown to be a stronger predictor of short-term price movements than volume alone.